

Q.1. For the memory locations whose physical address is specified by 1256Ah, give the logical address (in segment:offset form) for segments 1256h and 1240h.

Solution: let X be the offset in segment 1256h and Y the offset in segment 1240h. We have

$$\begin{aligned}1256A &= 12560h + X & \text{and} & & 1256Ah &= 12400h + Y \\X &= 1256Ah - 12560h & \text{and} & & Y &= 1256Ah - 12400h \\X &= \quad Ah & & & Y &= 16Ah \\1256Ah &= 1256:000A & \text{and} & & 1256Ah &= 1240:016A\end{aligned}$$

Q.2 A memory location has physical address 80FD2h. In what segment does it has offset BFD2h?

Solution:

$$\text{physical address} = \text{segment} \times 10h + \text{offset}$$

$$\begin{aligned}\text{Physical address} &= 80FD2 \text{ h} \\-\text{offset} &= BFD2 \text{ h}\end{aligned}$$

$$\text{Segment} \times 10h = 75000h$$

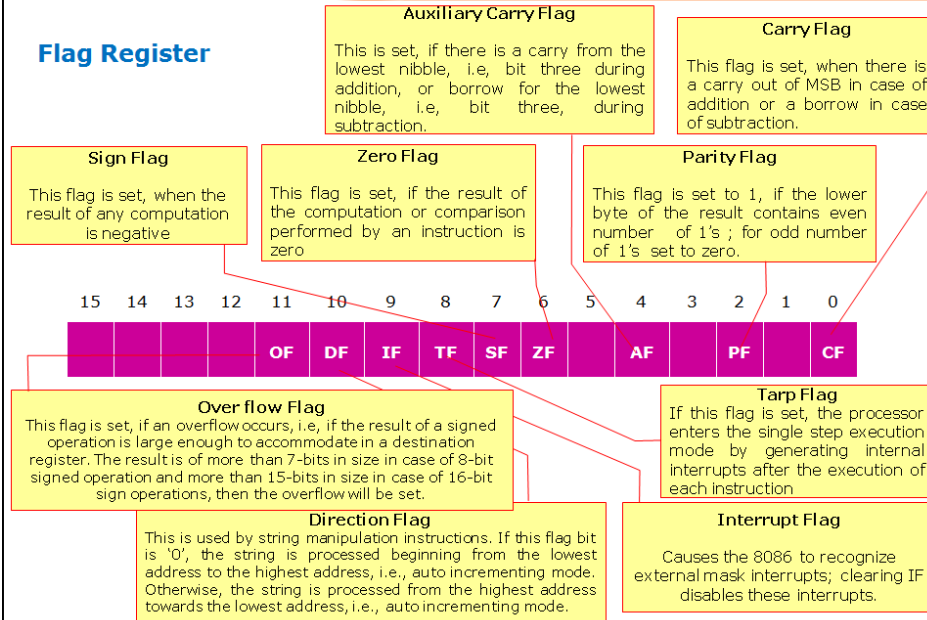
$$\text{Segment} = 75000h / 10h$$

$$\text{Segment} = 7500h$$

Architecture

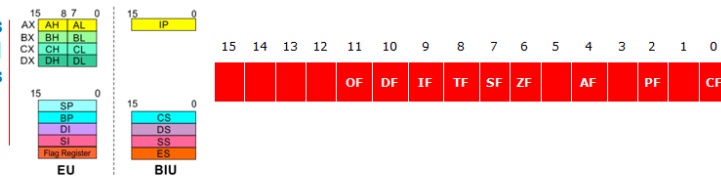
Execution Unit (EU)

Flag Register



Architecture

8086 registers categorized into 4 groups



Sl.No.	Type	Register width	Name of register
1	General purpose register	16 bit	AX, BX, CX, DX
		8 bit	AL, AH, BL, BH, CL, CH, DL, DH
2	Pointer register	16 bit	SP, BP
3	Index register	16 bit	SI, DI
4	Instruction Pointer	16 bit	IP
5	Segment register	16 bit	CS, DS, SS, ES
6	Flag (PSW)	16 bit	Flag register

Register	Name of the Register	Special Function
AX	16-bit Accumulator	Stores the 16-bit results of arithmetic and logic operations
AL	8-bit Accumulator	Stores the 8-bit results of arithmetic and logic operations
BX	Base register	Used to hold base value in base addressing mode to access memory data
CX	Count Register	Used to hold the count value in SHIFT, ROTATE and LOOP instructions
DX	Data Register	Used to hold data for multiplication and division operations
SP	Stack Pointer	Used to hold the offset address of top stack memory
BP	Base Pointer	Used to hold the base value in base addressing using SS register to access data from stack memory
SI	Source Index	Used to hold index value of source operand (data) for string instructions
DI	Data Index	Used to hold the index value of destination operand (data) for string operations